

Re-feeding Is Not Replaying: Measuring Replay Noise in Counterfactual Token-Credit Estimation

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Abstract

Per-token counterfactual credit estimation asks which token in a language-model rollout caused the final answer to be right or wrong: cut the transcript at a pivot, substitute an alternative token, replay continuations, and compare outcomes. Published methods implement the replay by *re-feeding* the transcript prefix as a fresh prompt, assuming the re-fed prefix reproduces the state the model passed through during generation. We measure what that assumption costs on a stock inference engine, with a three-pass design: continuations resumed from the verified decode-time KV state, an identical second exact pass serving as a replica noise floor, and a re-feed pass. Across six configurations and three models (including a GRPO-trained checkpoint), at low-margin decision tokens, re-feeding changes the credit estimate at rates 14–28 percentage points above the replica floor (7–21pp under a treatment-independent conditioning; problem-clustered $t = 2.9$ –6.4). Most changes are zero-boundary crossings of the quantized estimator rather than polarity reversals, and the perturbation is mean-zero, so averaged quantities are largely safe; but selection is not: a critical-token set chosen by thresholding $|\hat{A}_t|$ under re-feed overlaps the exact-resume selection at Jaccard 0.34–0.90, versus a 0.63–0.96 replica ceiling. A causal confirmation closes the loop: rerunning the harness under vLLM’s batch-invariant kernels makes all three passes identical on every measured channel, with both disagreement rates exactly zero. Even the floor is informative: replica passes disagree on 9–23% of eligible estimates, so single-sample credit measurements at decision tokens are unreliable under *any* replay. Settings were fixed in advance with a locked decision rule; cache hits underlying every “exact” claim are instrumented (100% hit rate, 3,434 pivots; 3,829 including the batch-invariant run); total compute was under \$10. We recommend that counterfactual credit studies resume decoder state or use batch-invariant kernels, and report a replica floor.

1 Introduction

Reinforcement learning with verifiable rewards has made per-token credit assignment a central measurement problem for language-model reasoning. Group relative policy optimization [14] and its descendants train on sequence-level rewards, so a growing line of work asks the finer-grained question: which individual tokens in a rollout caused success or failure? The standard answer is counterfactual replay. VinePPO [5] states the recipe’s premise directly: language environments “allow us to reset directly to any intermediate state simply by re-feeding the partial context.” Critical-token methods [9, 12] identify decisive tokens by regenerating continuations from token positions; causal credit constructions [6] and exact-counterfactual agent-credit methods [2] likewise reconstruct intermediate state from text.

The shared mechanical assumption is that re-feeding tokens $x_{1..t}$ puts the model in the state it was in when it generated token t . At the bit level it does not. Floating-point reduction order inside attention and matmul kernels depends on batch composition and on how a sequence is split between prefill and decode [4], so a re-fed prefix yields slightly different logits than the live decode state did. The working assumption in the credit literature is that these differences wash out of the final estimate. C3 [2] asserts the premise explicitly: that any decision point can be restored exactly from text. This paper measures that premise on the engines the literature actually runs on, and quantifies precisely what it costs, what it does not cost, and what eliminates it.

Contributions.

1. **A three-pass audit design with a replica floor.** We compare credit estimates from (i) continuations resumed from the decode-time KV state (cache hits instrumented per request: 100% verified across 3,434 pivots in the five stock-engine runs, 3,829 including the batch-invariant run), (ii) an identical second exact pass, and (iii) a re-feed pass. Pass (ii) converts “how often do estimates disagree?” into “how much disagreement is attributable to the re-feed procedure *beyond* replica noise?” The design runs on stock vLLM [7] with no kernel modifications.
2. **Quantification with disclosed decomposition.** At low-margin decision tokens, re-feeding changes the credit estimate 14–28pp above the floor across six configurations and three models (clustered $t = 2.9$ –6.4). We decompose the changes: most are zero-boundary crossings of the $1/K$ -quantized estimator; strict polarity reversals are 0.5–6.2% under re-feed versus 0–5.9% at the floor. The perturbation is mean-zero ($t = -1.2$), so aggregated estimates are largely unaffected.
3. **A downstream consequence: selection.** Threshold-based critical-token selection, the construction of Lin et al. [9] and Ruan et al. [12], selects a materially different token set under re-feed: Jaccard overlap with the exact-resume selection is 0.34–0.90 across configurations, against a replica ceiling of 0.63–0.96.
4. **A causal confirmation.** Rerunning the identical harness under vLLM’s batch-invariant kernels (VLLM_BATCH_INVARIANT=1, implementing 4) makes all three passes identical on every stored channel (credit estimates, greedy probe sequences, probe logprobs): both disagreement rates collapse to exactly zero and 100% of probes are bit-exact. In the tested configuration the effect is batch-variant kernel numerics, and the published fix eliminates it.
5. **The floor as a refinement.** Prior work established that batching numerics shift benchmark accuracy and flip greedy decoding [16, 15]. We measure the same hazard at the unit relevant to credit assignment: two bit-identical request batches disagree on 9–23% of eligible per-pivot estimates. Single-sample credit measurement at decision tokens is unreliable under any replay method.

Experimental settings (temperature, K , grading, pivot filter) and the decision rule were fixed before any GPU run, with one dated amendment (a difficulty screen) added after a smoke run and before any full-run results; the registration is self-attested (published in the repository, without an external timestamp) and the paper relies on it only as disclosure, not as proof. Total compute across both experimental campaigns was under \$10 on rented A100s. All per-pivot records, logs, the harness, and the analysis function that emits every number in this paper are public.

2 Related Work

Counterfactual token credit. VinePPO [5] builds Monte Carlo value estimates on the premise that re-feeding partial context resets the environment state. Lin et al. [9] identify critical tokens by rollouts from token positions; critical-token fine-tuning [12] forces alternative tokens and regenerates continuations. Khandoga et al. [6] construct causal per-token credit by masking reasoning spans and measuring answer-probability changes. C3 [2] shows exact counterfactuals beat approximate ones for credit assignment among cooperative LLM agents, at the granularity of whole agent actions, and premises its method on text-restoration being exact. All reconstruct intermediate state from text; none resume the original decoder state as a reference, and none report a replica control bounding the replay procedure’s own contribution. A contemporaneous survey of the area [18] repeats the premise: generating rollouts from any intermediate prefix is treated as trivially available, with no discussion of replay fidelity. Separately, tree-structured rollout methods [8] do resume live KV state at branch points, but as an efficiency mechanism during training, without comparing resume to re-feed or auditing exactness; our scope claim is therefore about credit *measurement* studies, not about all systems that branch from cached state. Chatzi et al. [1] study counterfactual token generation via shared sampling noise (Gumbel-max), an orthogonal notion of counterfactual that fixes randomness rather than internal state.

Inference nondeterminism and training mismatch. He and Thinking Machines Lab [4] identify batch invariance as the root cause of LLM inference nondeterminism and fix it with batch-invariant kernels; SGLang ships a comparable deterministic mode [13]. Yuan et al. [16] quantify accuracy swings and greedy argmax flips from batch-size numerics; Song et al. [15] show benchmark evaluation should not ignore sampling and numeric nondeterminism. On the training side, Zhong et al. [19] diagnose training-inference mismatch (rollout engine and trainer disagreeing on logprobs) as an independent cause of RL collapse, and Qi et al. [10] show FP16 precision largely removes that mismatch. These lines concern the serving endpoint and the training loop. Our victim is different: the *measurement methodology* of the credit-assignment literature, where the comparison pair is re-fed prefill state versus the original decode-time state, on stock engines. The lines compose: He et al. built the fix, the TIM line covers training, and this paper covers measurement, including a direct experimental confirmation that the batch-invariant fix closes the measurement gap (Section 5.4).

3 Method

3.1 Setup and estimator

For each problem we sample $N = 8$ trunk rollouts at temperature 0.7 with top- k logprobs recorded. The engine caches KV blocks as it decodes, so the state every rollout passed through remains resident in the prefix cache at block granularity. A *pivot* is a generated token whose top-1/top-2 logprob gap is below 1.0 (a low-margin decision token, the population credit methods target), restricted to positions within 2 tokens past a 16-token block boundary (block size read from the engine configuration and recorded), at most 5 per rollout, earliest first.

At pivot position t with sampled token a and best alternative a' , the credit estimate under a given replay method is

$$\hat{A}_t = \frac{1}{K} \sum_{k=1}^K R(x_{1..t-1}, a, c_k) - \frac{1}{K} \sum_{k=1}^K R(x_{1..t-1}, a', c'_k),$$

where c_k, c'_k are sampled continuations and R grades the completed solution by exact match on the GSM8K [3] final answer. Continuation seeds are a deterministic hash of (problem, rollout, pivot, arm, k), so all passes receive identical sampling randomness.

3.2 Three passes

Exact pass 1. All continuation requests for a problem are submitted while the trunk’s decode-time KV blocks are resident. Each request’s prompt cache-hits the state the rollout actually traversed. This is *instrumented*, not assumed: the engine’s per-request `num_cached_tokens` is recorded and checked against the block-aligned pivot position. Across all second-campaign runs, 100% of 3,829 pivots verified on both exact passes.

Exact pass 2 (replica floor). The identical request batch is submitted again, cache warm. Disagreement between passes 1 and 2 is the noise floor of the measurement itself. One impurity is disclosed: pass 1’s continuations also populated the cache, so pass 2’s requests can hit slightly more cached state than pass 1’s did, and its step-level batch composition differs accordingly. This makes the floor, if anything, slightly conservative as a baseline for the re-feed comparison.

Re-feed pass. The prefix cache is reset (the reset’s return value is checked; a silent failure would convert this pass into a third exact pass) and the identical batch is submitted. The trunk is freshly prefilled, exactly as in published credit pipelines. We attribute differences beyond the floor to *the re-feed procedure as actually run*: that includes both the recomputed prefix state and the changed step-level batch dynamics that heavy chunked prefill induces, because real re-feed pipelines batch their requests the same way.

The primary metric is the fraction of eligible pivots at which $\text{sign}(\hat{A}_t)$ under re-feed differs from exact pass 1, against the same quantity for exact pass 2. Eligibility is reported under two conditionings: the pre-fixed metric (any pass nonzero; shared denominator) and a treatment-independent conditioning ($\hat{A}_t^{\text{exact1}} \neq 0$), since the former lets the re-feed pass create its own eligibility (77 of 636 pooled eligible pivots are nonzero only under re-feed, versus 18 only under the replica). Sign changes are decomposed into zero-boundary crossings (one side zero) and strict polarity reversals. Each pass also issues one greedy 32-token probe per arm with logprobs recorded, giving a grading-free channel; probe deltas are per-pivot maxima over the two arms.

3.3 Why decode-time resume is the reference

The decode-time KV state is the state the model was actually in when it chose token t ; a counterfactual *do*-operation at t is defined relative to it. Resuming it requires no kernel modifications on paged-attention engines. Two scope notes. First, “exact” is exact to the 16-token block boundary: the trailing 0–2 tokens plus the forced token are computed as a short prefill chunk in both exact passes, so the resident KV is exact while the first post-pivot logits still traverse a fresh kernel path. Second, the replica floor measures the reference’s self-consistency, not its fidelity to the original decode-step logits; 61–94% of resumed probes reproduce the post-pivot logprob bit-exactly across configurations (versus 2.6–5.7% for re-feed), which validates self-consistency, and the batch-invariant confirmation (Section 5.4) closes the remaining gap by eliminating path dependence entirely.

3.4 Registration, campaigns, and disclosure

Temperature, K , grading, the pivot filter, and the decision rule were fixed in a registration document before any GPU run, with one dated amendment (a trunk-accuracy difficulty screen, blind to drift

Configuration	Model	n	Re-feed (pol.)	Floor (pol.)	Excess	t_{clust}	z
Main pooled ($t=.7$, K4)	Qwen2.5-7B-Inst	636	33.8% (3.9)	15.1% (1.1)	+18.7	5.51	8.61
$K=8$	Qwen2.5-7B-Inst	490	35.7% (4.9)	10.4% (1.0)	+25.3	6.36	9.68
$K=16$ (10 prob.)	Qwen2.5-7B-Inst	239	36.8% (5.9)	23.0% (5.9)	+13.8	2.87	3.71
Greedy cont. ($t=0$, K1) $^\diamond$	Qwen2.5-7B-Inst	189	40.2% (0.5)	13.8% (0.0)	+26.5	3.64	5.66
GRPO ckpt ($t=.7$, K4) $^\diamond$	SimpleRL-Zoo-7B	373	35.4% (3.8)	9.1% (1.1)	+26.3	5.50	8.66
Phi-4-mini ($t=.7$, K4)	Phi-4-mini	370	48.6% (6.2)	20.8% (3.0)	+27.8	3.47	7.79
Batch-invariant kernels	Qwen2.5-7B-Inst	106	0.0%	0.0%	0	—	—

Table 1: Estimate disagreement with exact resume on each run’s common eligible set: total rate, with strict polarity reversals in parentheses (percent). Excess in percentage points. t_{clust} is a problem-clustered paired t over per-problem rate differences; z is the (unclustered) McNemar statistic, reported for comparability. $^\diamond$ First campaign, prior to cache instrumentation; design identical (second-campaign instrumentation verified 100% cache-hit rates). In the greedy row, trunks are sampled at $t=0.7$ and only continuations are greedy. Under batch-invariant kernels all three passes are bitwise identical.

direction) added after a smoke run. One mechanism deviation is flagged: the registration described the exact arm as a snapshot-tool fork of the live KV state; the implementation uses the engine-native prefix-cache resume instead (functionally the same decode-time blocks, with no third-party component). The decomposition, conditioning, clustered, selection, and batch-invariant analyses are post-registration additions, labeled as such. The registration is published in the repository with an honest provenance note: it is self-attested, without an external timestamp. Two experimental campaigns are reported. The first campaign’s two pooled cohorts were later found to overlap on four problems (the screened cohort rescanned from index 0); all first-campaign pooled numbers in this paper are therefore deduplicated, and the second campaign separates cohort windows structurally and adds the cache instrumentation. Two first-campaign runs (greedy continuations; the GRPO checkpoint) are internally valid and reported as such, marked $^\diamond$ in Table 1.

4 Experimental Setup

Models. Qwen2.5-7B-Instruct [11]; Qwen-2.5-7B-SimpleRL-Zoo [17], a GRPO-trained checkpoint from the Qwen2.5-7B base; and Phi-4-mini-instruct (a distinct architecture and tokenizer family).

Engine. Stock vLLM 0.22.1 [7], bfloat16, prefix caching enabled. The batch-invariant confirmation sets `VLLM_BATCH_INVARIANT=1` on the same engine.

Hardware. One A100 SXM 80GB per run.

Data. GSM8K test split [3]; the main pooled configuration uses problems 0–19 plus 20 mixed-difficulty problems screened from index 20 upward (no overlap by construction).

Cost. Under \$10 total across both campaigns.

5 Results

5.1 Re-feed changes estimates far above the floor; most changes are zero-crossings

Table 1 gives the primary result. Re-feeding disagrees with the exact-resume estimate at 14–28 percentage points above the replica floor in every stock-engine configuration, with problem-clustered t between 2.9 and 6.4 (per-problem sign splits run 29-worse-versus-5 in the pooled configuration,

Pooled main configuration (40 non-overlapping problems, common eligible set $n=636$); red boxes = sign disagreement

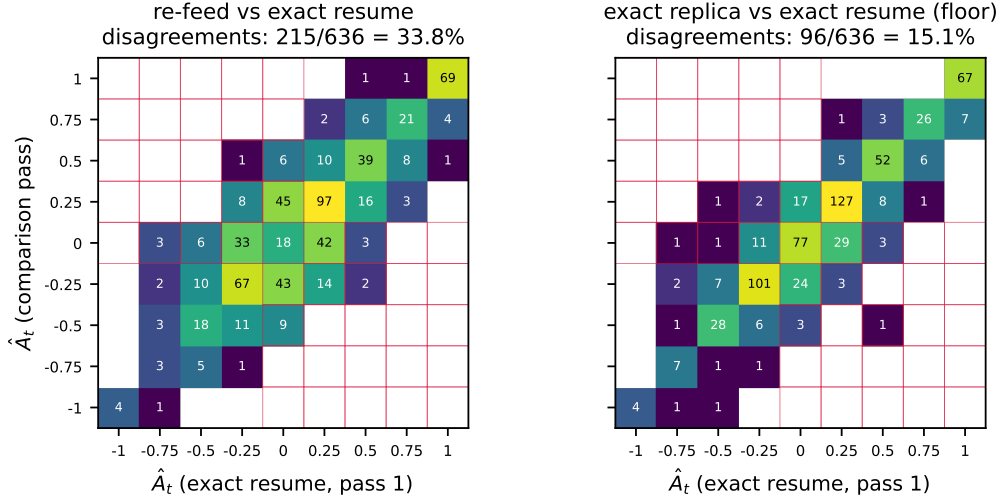


Figure 1: Joint distribution of $\hat{A}_t$ under exact resume (x-axis) and the comparison pass (y-axis), pooled main configuration. Red-outlined cells disagree in sign with exact pass 1; most disagreement mass sits in the zero row/column rather than in opposite-sign quadrants.

19-versus-0 at $K=8$, 16-versus-2 on the GRPO checkpoint). Under the treatment-independent conditioning the excess is smaller but decisive everywhere: +11.7pp pooled, +19.0 at $K=8$, +7.4 at $K=16$, +11.9 greedy, +17.1 GRPO, +21.1 Phi. There is no monotone K trend: $K=8$ is the strongest configuration.

The decomposition matters and is disclosed in the same table: strict polarity reversals (positive credit becoming negative or vice versa) are 0.5–6.2% under re-feed against 0–5.9% floors; most disagreements are the quantized estimator crossing zero. The perturbation is consistent with mean-zero (pooled bias -0.011 , $t = -1.23$, 95% CI $[-0.029, +0.007]$), so quantities that average many pivots are largely insulated. The effect is also flat in pivot margin (pooled: 34.2% below a 0.3 logprob gap versus 33.6% above; floors 12.0% and 16.9%). Figure 1 shows the full joint distribution; Figure 2 compares all configurations.

5.2 The downstream consequence: critical-token selection changes

Zero-mean noise still corrupts *selection*. Critical-token methods [9, 12] threshold per-token effects to choose which tokens to train on or analyze. Emulating that construction (select pivots with $|\hat{A}_t| \geq 0.5$), the set selected under re-feed overlaps the exact-resume selection at Jaccard 0.68 (pooled), 0.68 ($K=8$), 0.90 ($K=16$), 0.59 (greedy), 0.66 (GRPO), and 0.34 (Phi-4-mini), against replica ceilings of 0.84, 0.87, 0.96, 0.83, 0.88, and 0.63 respectively. Under batch-invariant kernels both overlaps are exactly 1.0. A third to two-thirds of the gap between a method’s selected token set and its own noise ceiling is attributable to the re-feed procedure.

To be precise about the mechanism: every selection disagreement we observe sits within one grading quantum ($1/K$) of the threshold, and the re-feed gap narrows as the estimator gets finer (Jaccard 0.68 at $K=4$ to 0.90 at $K=16$). This is not a separate phenomenon from the zero-crossing noise; it is that noise expressed at the cut. The point is that threshold selection at the K budgets the literature actually operates at lives exactly at that cut, and the replica ceilings show such selection is

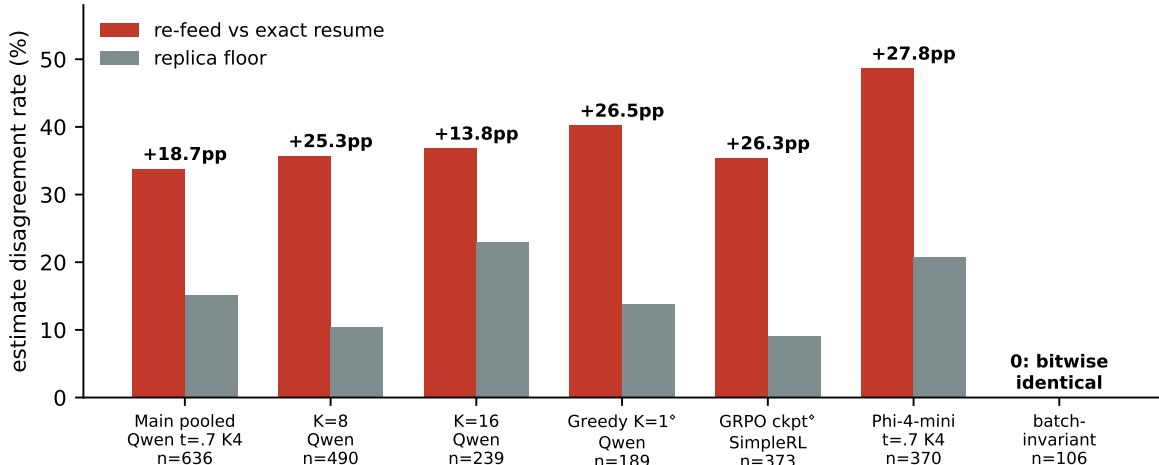


Figure 2: Disagreement rate of re-feed (red) versus the replica floor (gray) across all configurations. The excess survives the removal of sampling noise (greedy), grows on a second model family (Phi-4-mini), holds on a GRPO-trained checkpoint, and collapses to exactly zero under batch-invariant kernels.

substantially irreproducible under *any* replay at these budgets, with re-feeding consuming a further third to two-thirds of the remaining headroom.

5.3 Bit-level account

On the grading-free channel (greedy 32-token probes, per-pivot max over the two arms), resuming the decode-time state reproduces the first post-pivot token’s logprob exactly ($\delta = 0.0$) at 61–94% of pivots across configurations, while re-feeding does so at 2.6–5.7%; pooled 90th-percentile deltas are 3.7×10^{-3} (replica) versus 4.8×10^{-2} (re-feed). Probe sequences diverge within 32 greedy tokens at 22.8% of pooled pivots under re-feed versus 7.9% for the replica (Figure 3).

5.4 Causal confirmation: batch-invariant kernels eliminate the effect

If the measured excess is batch-variant kernel numerics and nothing else, then under kernels whose reduction order is independent of batch composition and prefill/decode split [4], all three passes should be identical. We reran the harness with `VLLM_BATCH_INVARIANT=1` on the same engine, model, and data (10 problems, 395 pivots, 106 eligible). The prediction held exactly: zero disagreements in both comparisons, 100% bit-exact probes in both comparisons, identical estimates on all 395 pivots, and critical-token Jaccard 1.0 (identity verified on every stored channel: estimates, greedy probe token sequences, and probe logprobs). This simultaneously closes the causal attribution and validates the instrument end-to-end: a harness artifact that manufactured drift would not measure exactly zero when the numerics are pinned.

5.5 The floor is also a finding

Two bit-identical request batches, submitted seconds apart with identical seeds, disagree on 9–23% of eligible per-pivot estimates and diverge on 2–15% of greedy probes. Prior work measured this hazard at the benchmark level [16, 15]; at the per-decision-token unit relevant to credit assignment it is large enough that single-sample estimates at low-margin tokens are uninterpretable under any

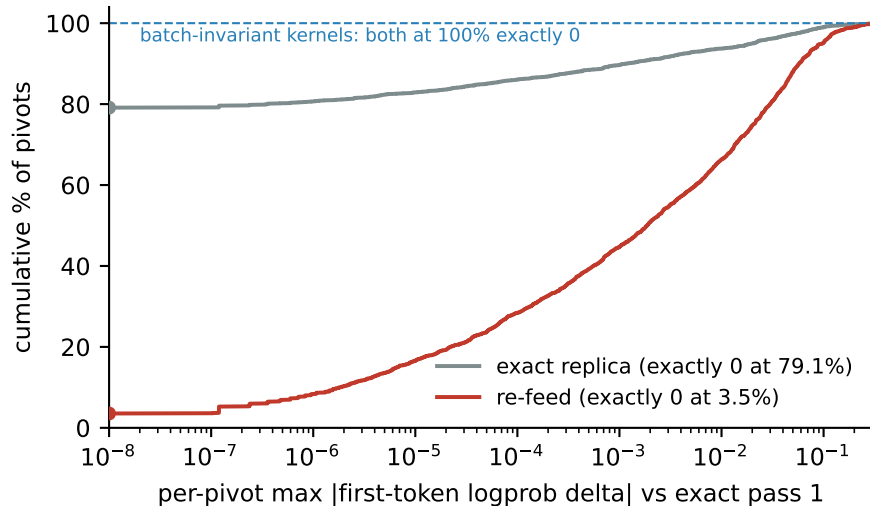


Figure 3: Cumulative distribution of the per-pivot maximum first-token logprob delta against exact pass 1, over the four instrumented second-campaign runs. The replica’s mass concentrates at exactly zero; re-feed’s essentially never does. Under batch-invariant kernels both curves degenerate to 100% at zero.

replay method on stock engines. Any credit-assignment study operating at this granularity needs a replica floor to be interpretable at all.

6 Discussion

What this does and does not show. It shows that the re-feed shortcut adds measurement noise far above the replica floor at decision tokens on stock engines, that this materially changes threshold-based token selection, and that batch-invariant kernels eliminate it entirely. It does not show that published aggregate conclusions are wrong: the perturbation is mean-zero, and methods that average over many tokens and rollouts are largely insulated. It does not claim novelty for the existence of batch numerics effects [4, 16]. One further scope note: same-process pipelines that re-feed with prefix caching enabled may silently cache-hit the decode-time blocks and thereby run the exact arm without knowing it; the indictment here applies to cache-cold re-feeds, which is what fresh-process and batched-analysis pipelines do.

Recommendations. (1) Run counterfactual replay on batch-invariant or deterministic kernels where available [4, 13]; our confirmation run shows this closes the gap exactly, at some throughput cost. (2) Otherwise, resume the decode-time state: on paged-attention engines this is engine-native within a process via the prefix cache, and durable session-snapshot tooling can extend it across processes. (3) In all cases report a replica floor; it costs one extra batch and is the difference between an interpretable and an uninterpretable single-pivot measurement. (4) Fix estimator settings in advance: the disagreement rate is sensitive to temperature, K , grading, and the pivot filter.

Conflict of interest. The author builds thaw, an open-source session-snapshot tool for vLLM whose premise (durable exact decoder state) is adjacent to recommendation (2). The experiments in this paper use stock vLLM only; the in-process resume used as the exact reference requires no

thaw component, and recommendation (1), a third party’s fix, is presented as the preferred path where available.

Limitations. One task family (GSM8K); one engine version (vLLM 0.22.1) and GPU type (A100); pivots are conditioned on low margin and block-boundary proximity, so all rates are statements about decision tokens, not all tokens; $K \leq 16$ with binary grading keeps the estimator coarse (the floor quantifies the consequence); the registration is self-attested; and two first-campaign runs predate the cache instrumentation (marked in Table 1). Passes always run in a fixed order (exact, replica, re-feed) and there is no re-feed-versus-re-feed replica, so the re-feed arm’s internal variance is unmeasured; we also do not isolate the recomputed-prefix mechanism from the changed batch dynamics with a serialized one-request-at-a-time control, which is why the treatment is defined as the re-feed procedure as actually run. The GRPO checkpoint was run only in the first campaign. The batch-invariant confirmation covers one configuration; replicating it across the full grid is mechanical.

7 Reproducibility

The registration (with its dated amendment and provenance note), the harness (one Python file against stock vLLM), every per-pivot record of all runs in both campaigns, run logs, and the analysis function that emits every number and figure in this paper are public in the thaw repository (<https://github.com/thaw-ai/thaw>: `benchmarks/`, `paper/refeed-drift/`). Total compute: under \$10 of rented A100 time. The first campaign’s cohort-overlap error is documented in the repository history rather than erased; pooled first-campaign numbers are deduplicated wherever cited.

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